

Implementation of Quantum Fourier Transform (QFT) Addition

```
In [3]: # !pip install qiskit_aer
%matplotlib inline

import sys
sys.path.append("../src")

from qiskit import QuantumRegister, QuantumCircuit, ClassicalRegister
from qiskit import transpile
import matplotlib.pyplot as plt
from qiskit.visualization import plot_histogram
from qiskit_aer import AerSimulator
import math
import numpy as np
from qiskit.synthesis.qft import synth_qft_full
from qiskit.circuit.library import DraperQFTAdder
```

The QFT adder implements

$$|a\rangle |b\rangle \mapsto |a\rangle |(a+b) \pmod{2^n}\rangle.$$

The circuit works in three steps:

1. Apply the QFT to the b-register to encode b in phases:

$$|a\rangle |b\rangle \mapsto |a\rangle \text{QFT}|b\rangle.$$

2. Use controlled phase rotations from a-register on the QFT-encoded b-register to add a in the phase domain:

$$|a\rangle \text{QFT}|b\rangle \mapsto |a\rangle \text{QFT}|a+b\rangle.$$

3. Apply the inverse QFT to recover the sum in the computational basis:

$$|a\rangle \text{QFT}|a+b\rangle \mapsto |a\rangle |(a+b) \pmod{2^n}\rangle$$

Setup in Qiskit - Starting point for reconciliation

```
In [58]: n = 3 # number of qubits

a_val = 2
b_val = 3

adder = DraperQFTAdder(num_state_qubits=n, kind='fixed') # modulo 2^n addition

qc = QuantumCircuit(2*n)

# Load a and b
for i in range(n):
    if (a_val >> i) & 1:
        qc.x(i)
```

```

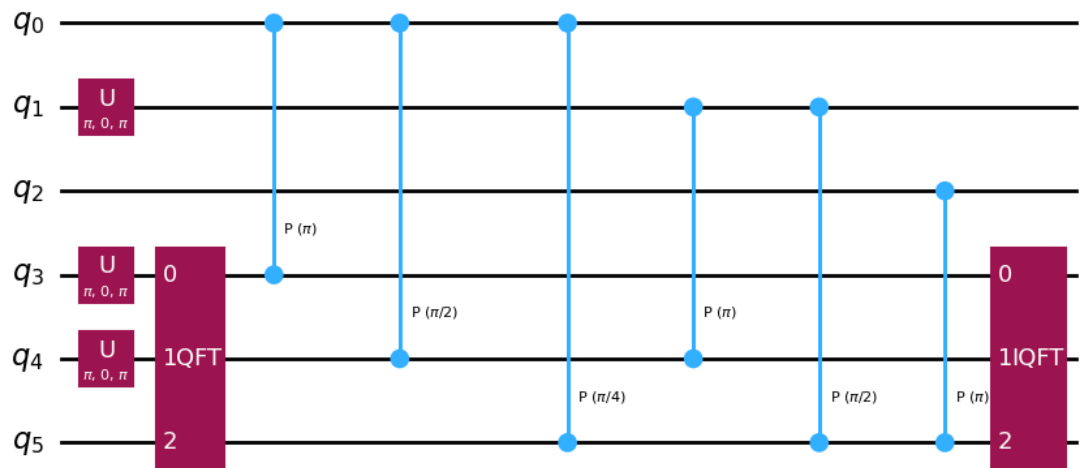
    if (b_val >> i) & 1:
        qc.x(n + i)

qc.append(adder, range(2*n))

qc.decompose().decompose().draw('mpl')

```

Out[58]:



Individual Gate implementation of the QFT Addition

```

In [59]: def qft_gate_3():
    qc = QuantumCircuit(3, name="QFT3")

    qc.h(2)
    qc.cp(np.pi/2, 2, 1)
    qc.cp(np.pi/4, 2, 0)

    qc.h(1)
    qc.cp(np.pi/2, 1, 0)

    qc.h(0)

    #qc.swap(0, 2)

    return qc.to_gate()

```

```

In [61]: n = 3 # number of qubits

a_val = 2
b_val = 3

a = QuantumRegister(n, 'a')
b = QuantumRegister(n, 'b')
creg = ClassicalRegister(len(b), 'creg')

qc = QuantumCircuit(a, b, creg)

# Prepare the circuit
for i in range(n):
    if (a_val >> i) & 1:

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        qc.x(a[i])
    if (b_val >> i) & 1:
        qc.x(b[i])

qc.barrier()

# Define QFT and IQFT (inverse QFT)
qft3 = qft_gate_3()
iqft3 = qft3.inverse()

# Apply QFT
qc.append(qft3, b)

# Adding a to b (correct)
for i in range(n):
    for k in range(n - i):
        angle = np.pi / 2**k
        qc.cp(angle, a[i], b[i + k])
    qc.barrier()

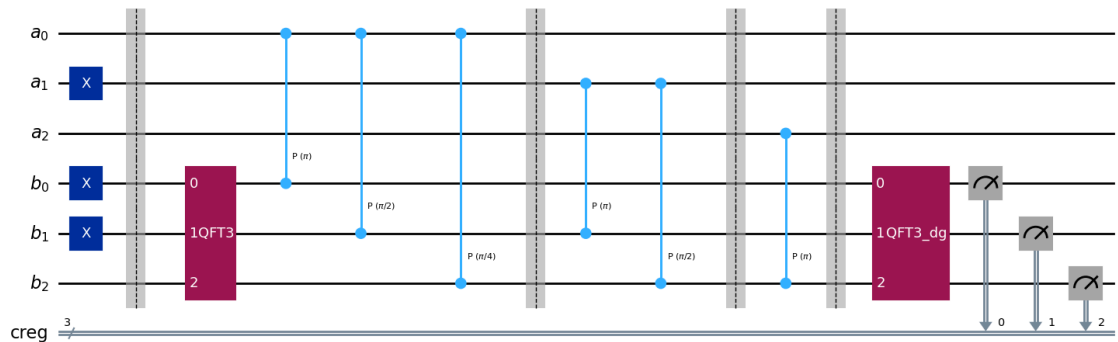
# Apply IQFT
qc.append(iqft3, b)

# Measure results
qc.measure(b, creg)

qc.draw('mpl')

```

Out[61]:



```

In [62]: sim = AerSimulator(method="matrix_product_state")

tqc = transpile(qc, sim)

nbr_shots = 2**12
result = sim.run(tqc, shots=nbr_shots).result()
counts = result.get_counts()

counts

```

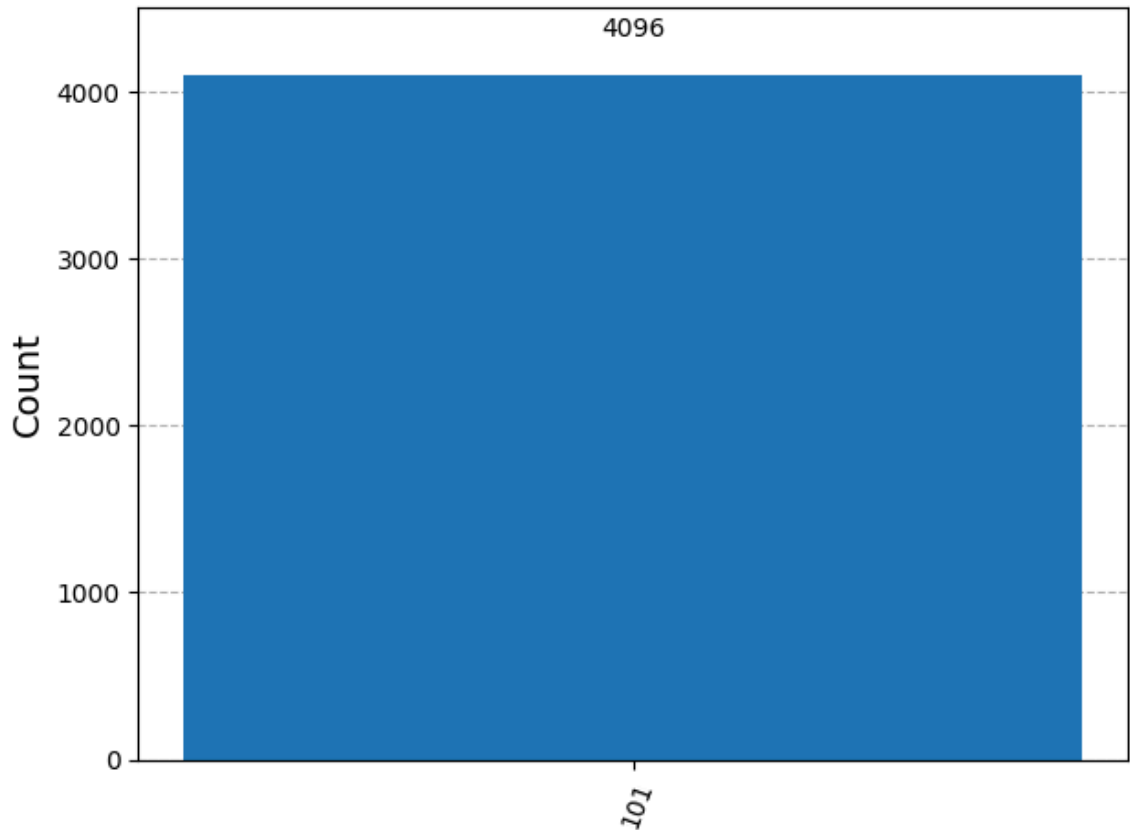
Out[62]: {'101': 4096}

```

In [63]: plot_histogram(counts)

```

Out[63]:



Expected result is

$$|2\rangle|3\rangle \mapsto |2\rangle|(2+3) \pmod{2^3}\rangle = |2\rangle|5 \pmod{2^3}\rangle = |2\rangle|5\rangle = |010\rangle|101\rangle$$

which got confirmed.

Wavefunction / Dirac Notation

Initial circuit

Note that the qubit order in a circuit in Qiskit $|q0, q1, q2\rangle$ is different from the textbook notation $|q2, q1, q0\rangle$.

The qubit on the bottom of the circuit is the MSB and the upper qubit is the LSB. We will use the textbook notation here.

We start with qubits a and b in the initial state $|000\rangle$

$$|\psi_0\rangle = |000\rangle_a \otimes |000\rangle_b$$

Set $a = 2$ and $b = 3$

$$|\psi_1\rangle = |010\rangle_a \otimes |011\rangle_b$$

Apply QFT

For our example with 3 qubits for each a and b and $N = 2^n = 2^3 = 8$ and $b = 3$ applying

$$\text{QFT}_N |x\rangle = \bigotimes_{j=0}^{n-1} \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i x / 2^{j+1}} |1\rangle \right)_{q_j}$$

we get

$$\text{QFT}_8 |3\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/2} |1\rangle \right)_{b_0} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/4} |1\rangle \right)_{b_1} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/8} |1\rangle \right)_{b_2}$$



Our full state becomes

$$|\psi_2\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/2} |1\rangle \right)_{b_0} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/4} |1\rangle \right)_{b_1} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/8} |1\rangle \right)_{b_2}$$



Apply controlled phase gates CR_1, CR_2, CR_3 from a_0 to b_0, b_1, b_2

With phases $e^{2\pi i / 2^k}$

- $b_0 : \theta = 2\pi i / 2^1 = 2\pi i / 2$
- $b_1 : \theta = 2\pi i / 2^2 = 2\pi i / 4$
- $b_2 : \theta = 2\pi i / 2^3 = 2\pi i / 8$

We start with the first one. The controlled phase gate R_1 applies $e^{i\pi} = e^{2\pi i \cdot \frac{1}{2}}$ if both a_0 and b_0 are in $|1\rangle_{a_0}$ and $|1\rangle_{b_0}$ state.

But as in this specific example $a_0 = |0\rangle$, the phase gate has no effect and the wave function remains unchanged.

$$|\psi_3\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/2} |1\rangle \right)_{b_0} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/4} |1\rangle \right)_{b_1} \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 3/8} |1\rangle \right)_{b_2}$$

The same is true for the next 2 controlled phase gates.



Apply controlled phase gates CR_1, CR_2 from a_1 to b_1, b_2

We apply phases θ in $e^{i\theta}$

- $b_1 : \theta = 2\pi / 2^1 = 2\pi / 2 = 2\pi / 2$
- $b_2 : \theta = 2\pi / 2^2 = 2\pi / 4 = 2\pi / 4$ to b_2

as $a_1 = |1\rangle$

$$|\psi_4\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 3/2} |1\rangle)_{b_0} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 2/4} e^{2\pi i 3/4} |1\rangle)_{b_1} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 1/4} e^{2\pi i 3/4} e^{2\pi i 2/4} |1\rangle)_{b_2}$$

Apply controlled phase gates CR_1 from a_2 to b_2

With phases

- $b_2 : \theta = 2\pi / 2^1 = 2\pi / 2 = 2\pi 2/4$

has again no effect since $a_2 = 0$ in our specific example, and the final result is therefore

$$|\psi_5\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 3/2} |1\rangle)_{b_0} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 2/4} e^{2\pi i 3/4} |1\rangle)_{b_1} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 1/4} e^{2\pi i 3/4} e^{2\pi i 2/4} |1\rangle)_{b_2}$$

Note that $e^{2\pi i 3/2} = e^{2\pi i 5/2}$, since

$$\frac{3}{2} \equiv \frac{1}{2} \pmod{1}, \quad \frac{5}{2} \equiv \frac{1}{2} \pmod{1}.$$

Imagine a circle and you can go half circles (a full circle is 2π). Whether you go 3 half-circles or 5 half-circles, you will always end up at π .

Now we can write

$$|\psi_5\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 5/2} |1\rangle)_{b_0} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 2/4} e^{2\pi i 3/4} |1\rangle)_{b_1} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 1/4} e^{2\pi i 3/4} e^{2\pi i 2/4} |1\rangle)_{b_2}$$

$$|\psi_5\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (2+3)/2} |1\rangle)_{b_0} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (2+3)/4} |1\rangle)_{b_1} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i (2+3)/4} |1\rangle)_{b_2}$$

and further simplify to

$$|\psi_5\rangle = |010\rangle_a \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 5/2} |1\rangle)_{b_0} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 5/4} |1\rangle)_{b_1} \otimes \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i 5/4} |1\rangle)_{b_2}$$

Now we see the result $(2+3)=5$ exactly in the exponent.

Applying the Inverse QFT yields as expected the following result in the computational basis:

$$|\psi_6\rangle = |010\rangle_a \otimes |101\rangle_b$$

$$|\psi_6\rangle = |2\rangle_a \otimes |5\rangle_b$$

Next steps:

- Extend the algorithm for a QFT modular adder
- write the logic for the general case $(a+b)$